

Intelligent Farmland Agent: Pitch Deck & Business Model Canvas

Project Name: Intelligent Farmland Agent (AgriTech)

Team: FuTech (Vijay Sarathi R.S, Sanjai S, Akhil T.T, Sreevidhya A)

Tagline: Asset intelligence for farmland — auditable, time-anchored, and AI-driven.

Slide 1: Title Slide

- **Visual:** High-quality image of a farm overlaid with digital data points/grids.
- **Heading:** Intelligent Farmland Agent
- **Sub-heading:** Asset intelligence for farmland — auditable, time-anchored, and AI-driven.
- **Footer:** Team FuTech

Slide 2: The Problem

"Farming is high-risk, but data shouldn't be."

- **For Farmers:** Crop health monitoring is manual, labor-intensive, and reactive. Unexpected weather events devastate yields without warning.
- **For Insurers & Lenders:** Verifying claims relies on outdated methods or expensive site visits. There is a lack of "truth" regarding historical field conditions.
- **For Regulators:** Compliance reporting is tedious and often lacks data-backed evidence.

Slide 3: The Solution

Intelligent Farmland Agent

An AI-powered web application that serves as a central intelligence hub for farmland assets.

- **Satellite Intelligence:** Uses NASA/Satellite imagery to calculate vegetation health (NDVI).
- **Weather Risk Engines:** Integrates historical and forecast data to compute risk scores.
- **Generative AI (Gemini):** Contextualizes complex data into actionable human-readable insights.
- **Automated Reporting:** Generates auditable PDF reports for insurance, banking, and compliance.

Slide 4: How It Works (The Tech Stack)

- **Input:** Farmer registers a field (GPS coordinates, crop type, acres).
- **Process:**
 1. **Satellite Service:** Fetches imagery & computes NDVI.
 2. **Weather Service:** Analyzes historical patterns & forecasts risks.
 3. **AI Analyzer:** Google Gemini synthesizes this data to explain *why* crops are stressed.

- **Output:** Real-time dashboards and downloadable, time-stamped PDF reports.

Slide 5: Market Opportunity

- **Precision Agriculture:** A rapidly growing market projected to reach billions by 2030.
- **Climate Resilience:** Increasing extreme weather events create a desperate need for better risk management tools.
- **Agri-Insurance:** Insurers are actively seeking remote, data-driven methods to automate payouts and detect fraud.

Slide 6: Competitive Advantage

- **Generative AI Integration:** Unlike standard dashboards that just show graphs, our agent *explains* the data using Google Gemini.
- **Resilience:** Multi-API fallbacks ensure data availability even when primary sources fail.
- **Auditable Evidence:** We focus on generating verified "evidence" (PDFs) suitable for third-party auditing, not just operational monitoring.

Slide 7: Business Model Canvas (The 9 Building Blocks)

(Note for presentation: This can be displayed as a visual 9-box grid on the slide)

1. Customer Segments

- **Direct Users:** Small to Medium-sized farm owners and farm managers.
- **Enterprise:** Crop Insurance Companies (Risk Assessors & Adjusters).
- **Financial & Govt:** Agri-lenders verifying asset health; Agricultural regulators.

2. Value Propositions

- **Risk Reduction:** Early warning systems combining weather and crop health.
- **Verification & Trust:** Third-party auditable, time-anchored records (PDFs) for claims.
- **Actionable Insights:** AI translates raw satellite/weather data into plain English.
- **Time Saving:** Automating physical "scouting" via remote satellite monitoring.

3. Channels

- **Web Platform:** Direct sign-up for the SaaS web application.
- **API Portal:** Developer documentation for enterprise B2B integrations.
- **Partnerships:** Bundled with agricultural insurance policies or farming equipment.

4. Customer Relationships

- **Automated Service:** Self-service dashboard for farmers with automated alerts.
- **API Integration Support:** Dedicated technical support for enterprise clients.
- **Community & Education:** Resources on interpreting NDVI and weather risk data.

5. Revenue Streams

- **Subscription (SaaS):** Tiered monthly fees for continuous monitoring & AI alerts.
- **Enterprise API Licensing:** Volume-based pricing for insurers accessing backend data.

- **Pay-Per-Report:** One-off transaction fees for generating certified historical audit reports.

6. Key Resources

- **Data APIs:** Google Gemini, NASA/Satellite imagery, Weather Services (OpenWeatherMap).
- **Intellectual Property:** Proprietary algorithms for risk scoring and AI prompt injection.
- **Infrastructure:** Cloud hosting (Render, Docker) and databases.

7. Key Activities

- **Data Aggregation:** Reliable fetching, fallback routing, and cleaning of data.
- **AI Tuning:** Optimizing Gemini prompts for hallucination-free agricultural context.
- **Platform Development:** Maintaining the Flask app, UI, and backend services.
- **Report Generation:** Ensuring PDF accuracy, formatting, and secure storage.

8. Key Partnerships

- **Data Providers:** NASA, ESA, and global weather data providers.
- **Technology Partners:** Google Cloud (Gemini AI), hosting providers.
- **Institutional Partners:** Pilot programs with crop insurance firms and agricultural banks.

9. Cost Structure

- **API Costs:** Token usage for Gemini, premium API calls for high-res weather/satellite data.
- **Infrastructure:** Cloud hosting, database storage, and compute instances.
- **Development & Maintenance:** Salaries for engineering and platform maintenance.
- **Sales & Marketing:** Customer acquisition costs for B2C and B2B segments.

Slide 8: Feature Roadmap

Phase 1: MVP (Completed for Hackathon)

- Field Registration, NDVI Calculation, Weather Risk Analysis, Gemini AI Insights, PDF Reports.

Phase 2: Enhanced Intelligence (Next Steps)

- **Historical Trends:** Comparative analysis of this season vs. the last 5 years.
- **Pest Prediction:** Using hyper-local weather patterns to predict outbreaks.
- **Alerting System:** SMS/WhatsApp push notifications for critical health drops.

Phase 3: Ecosystem Integration (Future Vision)

- **IoT Integration:** Ingesting data from ground sensors (soil moisture) to complement satellites.
- **Agri-Marketplace:** Connecting farmers with suppliers based on AI-identified deficiencies.

Slide 9: The Team

Team FuTech

- Vijay Sarathi R.S

- **Sanjai S**
- **Akhil T.T**
- **Sreevidhya A**
- *Closing Note:* Thank you for your time. Open for questions!